

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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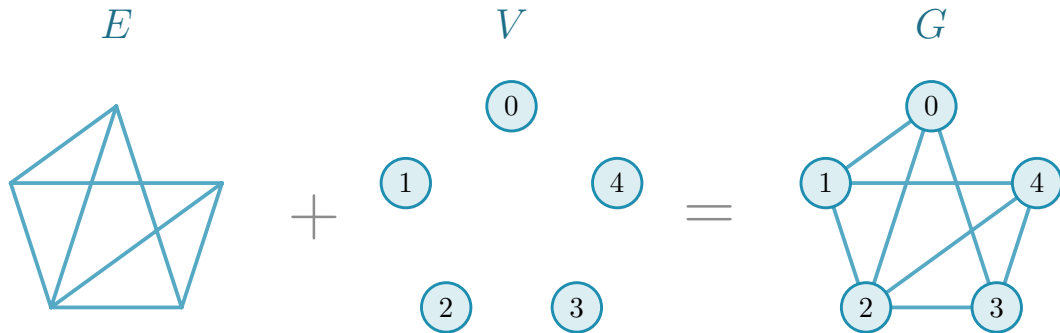
Maximising edge density in graphs
under connectivity constraints



translation

networks with **many links** and **low connectivity**

Graph

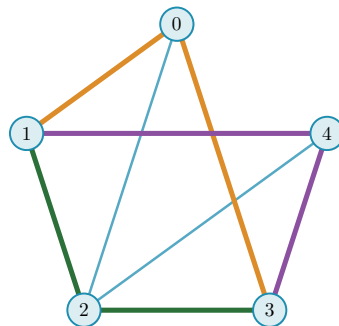


Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$ = the largest number of u - v routes that share **no edge**: one colour each

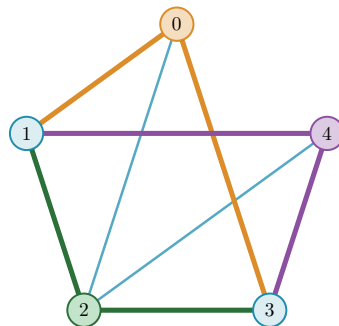


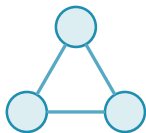
The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \kappa_G^{\max} \leq m-1 \right\}$$

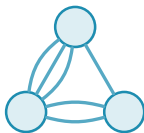
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

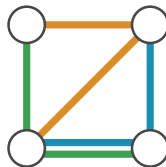




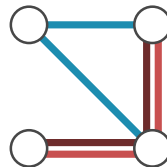
simple graph



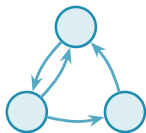
multigraph



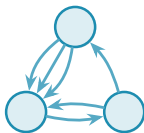
hypergraph



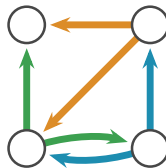
multihypergraph



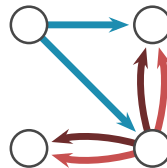
simple digraph



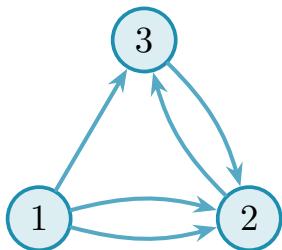
directed multigraph



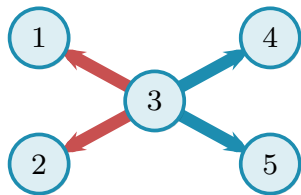
directed hypergraph



directed multihypergraph



$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

directed multigraph

max flow
over every pair

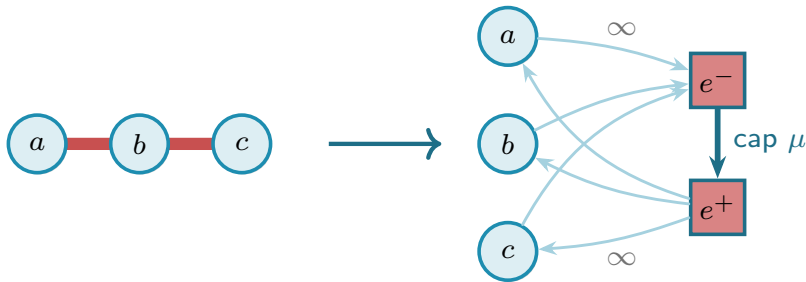
$$\lambda_G^{\max} = 3$$

OUTPUT

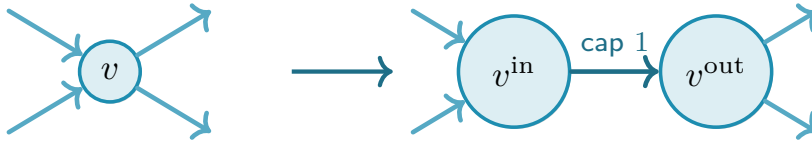
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



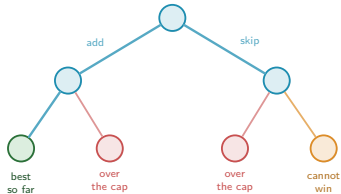
$$\kappa \rightarrow \lambda$$



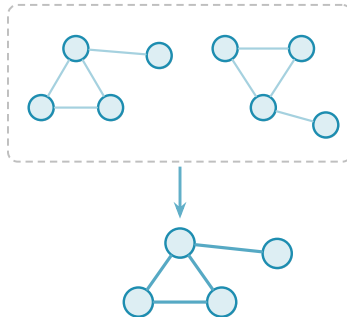
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

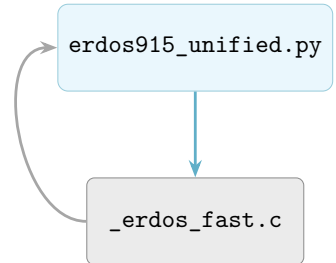
pruning



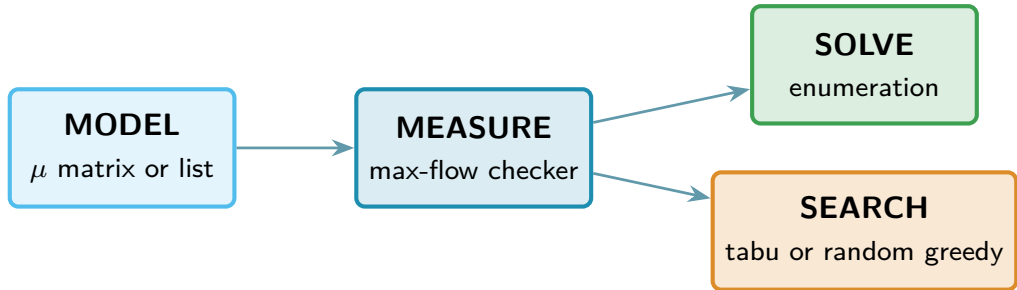
generating search



C: faster max flow

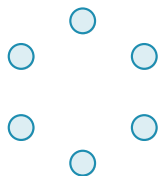


Solving versus searching

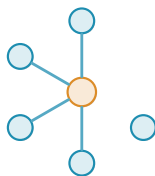


The search: one number per graph

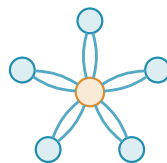
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



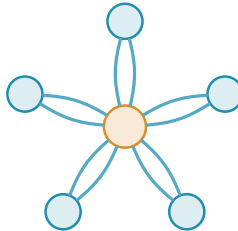
$$\Phi = -10 \text{ at } m = 3$$

Extremal families the search rediscovers



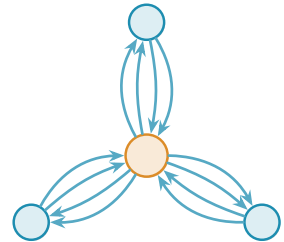
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$

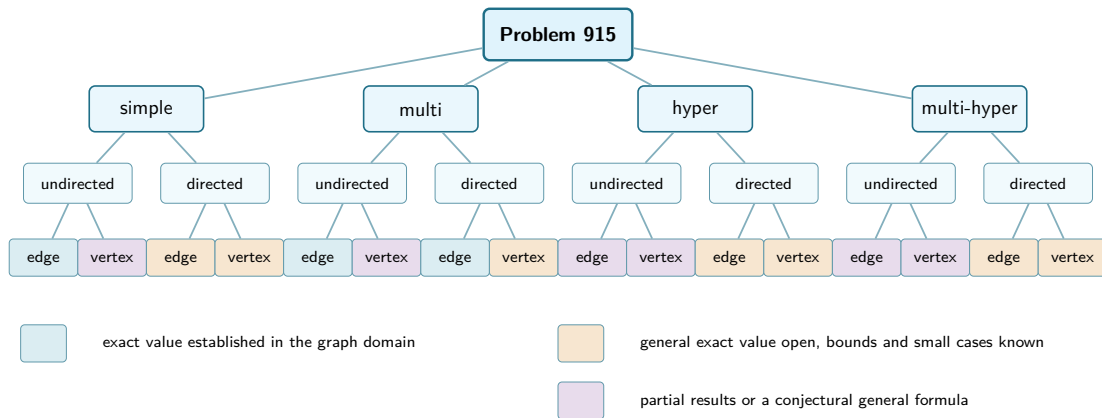


double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

Status of the sixteen variants



Thank you

`github.com/louisvandenbruwaene/thesis`

Backup: the sixteen variants

HOW TO READ THE NEXT SIXTEEN FRAMES

Each frame fixes one small size and gives the two bounds there as numbers, rather than a rule for general n and m .

PROVED

established at that size, upper bound and lower bound alike

EXTREMAL

the drawing is a witness meeting the upper bound exactly, so every vertex and every edge on it can be counted

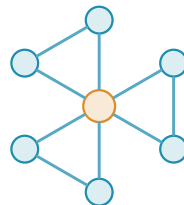
Simple graph · edge

$$\ell_m(n)$$

UPPER 9 **PROVED**

LOWER 9 **PROVED**

- the drawing is a hub joined to six leaves, with a perfect matching added among the leaves: seven vertices, nine edges



$n = 7, m = 3$ **EXTREMAL**

COMPUTED Exact at every size with $n \geq m$, by Mader's theorem via Gomory–Hu.

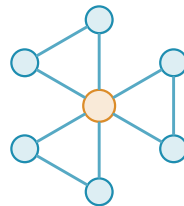
Simple graph · vertex

$$k_m(n)$$

UPPER 9 **PROVED**

LOWER 9 **PROVED**

- the same graph as the edge problem, because the vertex and edge values coincide while m is at most four
- from $m = 5$ the two part company, and the general value is open



$n = 7, m = 3$ **EXTREMAL**

COMPUTED Exact for $m \leq 4$ at every n . At $m = 5$ the two agree up to $n = 13$ and first differ at $n = 14$.

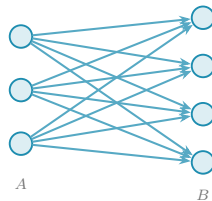
Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

UPPER 12 **PROVED**

LOWER 12 **PROVED**

- the drawing sends every arc from three tails to four heads: seven vertices, twelve arcs
- a hub bidirected to six spokes also holds twelve, so at this size the two branches tie and the witness is not unique



$n = 7, m = 2$ **EXTREMAL**

COMPUTED Exact at every n for $m = 2$. For $m \geq 3$ only the growth rate is settled, and the exact values stay open.

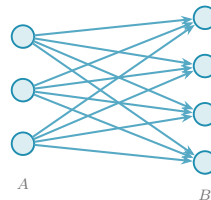
Simple digraph · vertex

$$k_m^{\text{dir}}(n)$$

UPPER 12 **PROVED**

LOWER 12 **PROVED**

- the same digraph as the arc problem: at $m = 2$ the two separations agree at every size



$n = 7, m = 2$ **EXTREMAL**

COMPUTED Exact at every n for $m = 2$. Whether arc and vertex agree for $m \geq 3$ is open.

Multigraph · edge

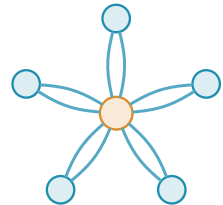
$$L_m(n)$$

UPPER 10 **PROVED**

LOWER 10 **PROVED**

- the drawing is a hub joined to five leaves with every edge doubled: six vertices, ten edges
- any tree does as well as the star, since only the cut count matters

COMPUTED Exact at every size, by a count on a single cut.



$n = 6, m = 3$ **EXTREMAL**

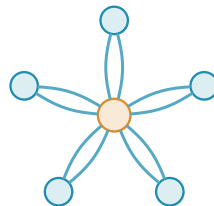
$$K_m(n)$$

UPPER 10 **PROVED**

LOWER 10 **PROVED**

- the same graph as the edge problem, because the vertex and edge values coincide at $m = 3$
- from $m = 4$ the general value is open, and parallel copies then buy more than the simple problem allows

COMPUTED Exact at $m = 2$ and $m = 3$ for every n . Beyond that only single sizes are settled, such as 14 at $n = 4$, $m = 5$ by exhaustion.



$n = 6, m = 3$ **EXTREMAL**

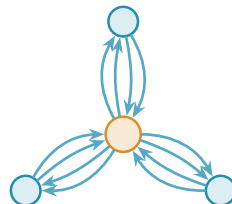
Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

UPPER 12 **PROVED**

LOWER 12 **PROVED**

- the drawing is a hub bidirected to three spokes with every arc doubled: four vertices, twelve arcs



$n = 4, m = 3$ **EXTREMAL**

COMPUTED Exact at every size, and one threshold-free optimisation settles all of them at once.

Directed multigraph · vertex

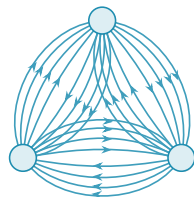
$$K_m^{\text{dir}}(n)$$

UPPER 24 **PROVED**

LOWER 24 **PROVED**

- every one of the six directions carries four copies: three vertices, twenty-four arcs, and it is the only shape that reaches the maximum
- four is the cap, not five, because an ordered pair also picks up one route through the third vertex, which brings it to five in total
- the arc extremiser does not reach this: twenty arcs against twenty-four

COMPUTED Enumerating every directed multigraph on three vertices leaves this one isomorphism class. Larger sizes stay open.



$n = 3, m = 6$ **EXTREMAL**

Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

UPPER 4 **PROVED**

LOWER 4 **PROVED**

- the drawing is a hub and four disjoint pairs of leaves, one hyperedge per pair: nine vertices, four hyperedges
- a hyperedge spans three vertices but still serves only one route, which is why the count divides by two



$$n = 9, r = 3, m = 2$$

EXTREMAL

COMPUTED Exact whenever there are enough distinct hyperedges to go round, which holds here. It can fail at the smallest sizes, such as $n = r = m = 3$.

Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

UPPER 4 **PROVED**

LOWER 4 **PROVED**

- the same hypergraph as the edge problem: at $m = 2$ the two separations agree for every uniformity
- at $m = 2$ a hypergraph is feasible exactly when its incidence graph is a forest, and this one is a tree



$n = 9, r = 3, m = 2$

EXTREMAL

COMPUTED Exact at $m = 2$ and $m = 3$. From $m = 4$ the general value is open, since an edge bound does not carry over to vertex separation.

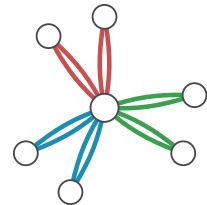
Multihypergraph · edge

$$L_m^{(r)}(n)$$

UPPER 6 **PROVED**

LOWER 6 **PROVED**

- the drawing is a hub and three disjoint pairs of leaves, each hyperedge taken twice: seven vertices, six hyperedges
- repetition leaves the bound where it was but changes which shapes reach it



$$n = 7, r = 3, m = 3$$

EXTREMAL

COMPUTED Exact whenever the pairs divide the leaves evenly, which is the case here.

Multihypergraph · vertex

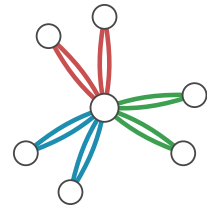
$$K_m^{(r)}(n)$$

UPPER 6 **PROVED**

LOWER 6 **PROVED**

- the same hypergraph as the edge problem
- repetition buys nothing at $m = 2$, where two copies of one hyperedge are already two separated routes, but it does buy something from $m = 3$

COMPUTED Exact at $m = 2$ and $m = 3$. Larger thresholds are open.



$n = 7, r = 3, m = 3$

EXTREMAL

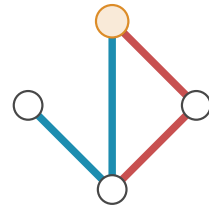
Directed simple hypergraph · arc

$$\ell_m^{\text{dir}(r)}(n)$$

UPPER 8 **PROVED**

LOWER 8 **PROVED**

- every vertex is the tail of two hyperedges, each carrying two of the other three as heads: four vertices, eight hyperedges
- one tail is drawn in full, in orange, and the other three repeat it around the square



$$n = 4, r = 3, m = 3$$

EXTREMAL

COMPUTED Deleting a vertex's own outgoing copies cuts it off, which is why the count per tail is capped. Larger sizes have a settled growth rate but open exact values.

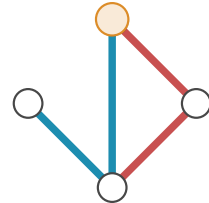
Directed simple hypergraph · vertex

$$k_m^{\text{dir}(r)}(n)$$

UPPER 8 **PROVED**

LOWER 8 **PROVED**

- the same hypergraph as the arc problem, which is feasible for both separations at once



$$n = 4, r = 3, m = 3$$

EXTREMAL

COMPUTED Whether the two separations keep the same exact value at larger sizes is open.

Directed multihypergraph · arc

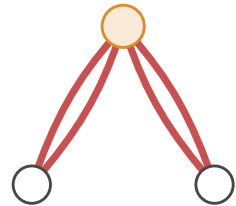
$$L_m^{\text{dir}(r)}(n)$$

UPPER 6 **PROVED**

LOWER 6 **PROVED**

- at this size a tail has only one possible hyperedge, the other two vertices, so it takes two copies of it: three vertices, six hyperedges
- repetition is what lifts the limit on distinct head sets, and it is the whole difference from the simple model

COMPUTED Forward, backward and general orientations share the same growth rate. The exact values at larger sizes are open.



$$n = 3, r = 3, m = 3$$

EXTREMAL

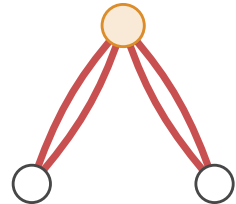
Directed multihypergraph · vertex

$$K_m^{\text{dir}(r)}(n)$$

UPPER 6 **PROVED**

LOWER 6 **PROVED**

- the same hypergraph as the arc problem



$$n = 3, r = 3, m = 3$$

EXTREMAL

COMPUTED Whether forward and general orientations agree once n passes r is open.